

| Input               | Process                                                     | Output                                                                    |
|---------------------|-------------------------------------------------------------|---------------------------------------------------------------------------|
| Principal amount    | Loop for 5 years                                            | Year, beginning balance, interest, and ending balance for each of 5 years |
| Interest Rate       | Compute interest =<br>beginning balance $\times$ rate       | Total accumulated interest after 5 years                                  |
| Number of years = 5 | Compute ending balance =<br>beginning balance +<br>interest |                                                                           |
|                     | Add interest to total<br>accumulated interest               |                                                                           |
|                     | Set beginning balance =<br>ending balance for next year     |                                                                           |

|      |                                                    |                                                            |
|------|----------------------------------------------------|------------------------------------------------------------|
| None | Set the first two numbers to<br>1 and 1            | The first 20 Fibonacci<br>numbers starting with 1 and<br>1 |
|      | Use a loop to get 20<br>numbers                    |                                                            |
|      | Add the last two numbers to<br>get the next number |                                                            |
|      | Move the numbers over                              |                                                            |

|                                                                 |                                         |                                                            |
|-----------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------|
| A text file that has each<br>employee's last name and<br>salary | Open the file and read each<br>line     | For each employee: show<br>last name, salary, and<br>bonus |
|                                                                 | Look at their salary                    | At the end: show the total of<br>all bonuses paid out      |
|                                                                 | Decide the bonus rate:                  |                                                            |
|                                                                 | \$100,000 and up $\rightarrow$ 20%      |                                                            |
|                                                                 | \$50,000 $\rightarrow$ 15%              |                                                            |
|                                                                 | Anything else $\rightarrow$ 10%         |                                                            |
|                                                                 | Compute bonus = salary $\times$<br>rate |                                                            |
|                                                                 | Add the bonus to a running<br>total     |                                                            |

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|  | Repeat until all employees have been processed |  |
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|-------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| A text file that has: item name, quantity, and price on each line | Read the file one line at a time                                     | For each line: show item, quantity, price, and extended price                          |
|                                                                   | Get the item, quantity, and price                                    | At the end: show total of all extended prices, number of orders, and the average order |
|                                                                   | Compute extended price = quantity $\times$ price                     |                                                                                        |
|                                                                   | Add extended price to a running total                                |                                                                                        |
|                                                                   | Add 1 to the order count                                             |                                                                                        |
|                                                                   | Compute average order = total extended price $\div$ number of orders |                                                                                        |

|                                                                                      |                                                    |                                                                    |
|--------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------|
| A text file with: Student last name, District code (I or O), Number of credits taken | Read the file one student at a time                | For each student: show last name, credits taken, and tuition owed  |
|                                                                                      | Check district code                                | At the end: show the total tuition owed and the number of students |
|                                                                                      | If I, cost per credit = 250                        |                                                                    |
|                                                                                      | If O, cost per credit = 500                        |                                                                    |
|                                                                                      | Compute tuition = credits $\times$ cost per credit |                                                                    |
|                                                                                      | Add tuition to a running total                     |                                                                    |
|                                                                                      | Add 1 to the student count                         |                                                                    |